# Synthetic Tabular Data for Fraud Detection

CTGAN vs. FinDiff: Distribution Fidelity, Ratio Control & Downstream Performance

RQ1 Improve or replace real data?

RQ2 Ratio forcing → distribution & performance?

RQ3 GAN vs. Diffusion — paradigm differences?

UCI Credit Card Fraud Dataset · XGBoost Downstream Classifier · Viva Presentation

INTRODUCTION

# Why Synthetic Data for Fraud Detection?

### Data Sensitivity

Real fraud transaction data cannot be shared freely limits reproducibility and research collaboration.

### Extreme Imbalance

Fraud rate < 0.2% in practice. Standard classifiers are biased towards the majority class.

### Evaluation Gap

Most synthetic data work measures statistical similarity, not downstream detection utility.

### What this study adds

- Evaluate synthetic data by downstream Fraud F1 & AUPRC, not just KS distance
- Explicitly manipulate class ratios and measure the effect on distribution and performance
- Compare GAN (CTGAN) vs Diffusion (FinDiff) under identical conditions
- Analyse double-penalisation: synthetic oversampling + scale_pos_weight

RELATED WORK

# Prior Work & Gaps

## Imbalance Handling

- **SMOTE: interpolation-based oversampling** (Chawla et al. 2002)
- **Cost-sensitive learning: scale_pos_weight in XGBoost** (Chen & Guestrin, 2016)
- **Limitation: struggle with high-dimensional feature dependencies**

## Tabular Generators

- **CTGAN: conditional GAN for mixed/imbalanced tabular data** (Xu et al. 2019)
- **Diffusion models: iterative denoising** (Ho et al. 2020; Song et al. 2021)
- **TabDDPM: diffusion for tabular data** (Kotelnikov et al. 2023)

## Key Gaps

- **Many studies use single fixed class ratio** (D'souza et al. (AAAI, 2025),(Panagiotou, Roy & Ntoutsi (arXiv, 2024),(Kiran & Saravana Kumar (IEEE, 2023)
- **Downstream utility rarely evaluated directly**
- **Diffusion under extreme imbalance understudied** (Schreyer et al. (ICAIF, 2024), Zheng et al. (arXiv, 2024)
- **Unexplored effects of varying fraud ratios during data generations**

→ **This study directly addresses all three gaps**

PROBLEM DEFINITION

# Experimental Setup

**284,807**
Total transactions

**492**
Fraud samples

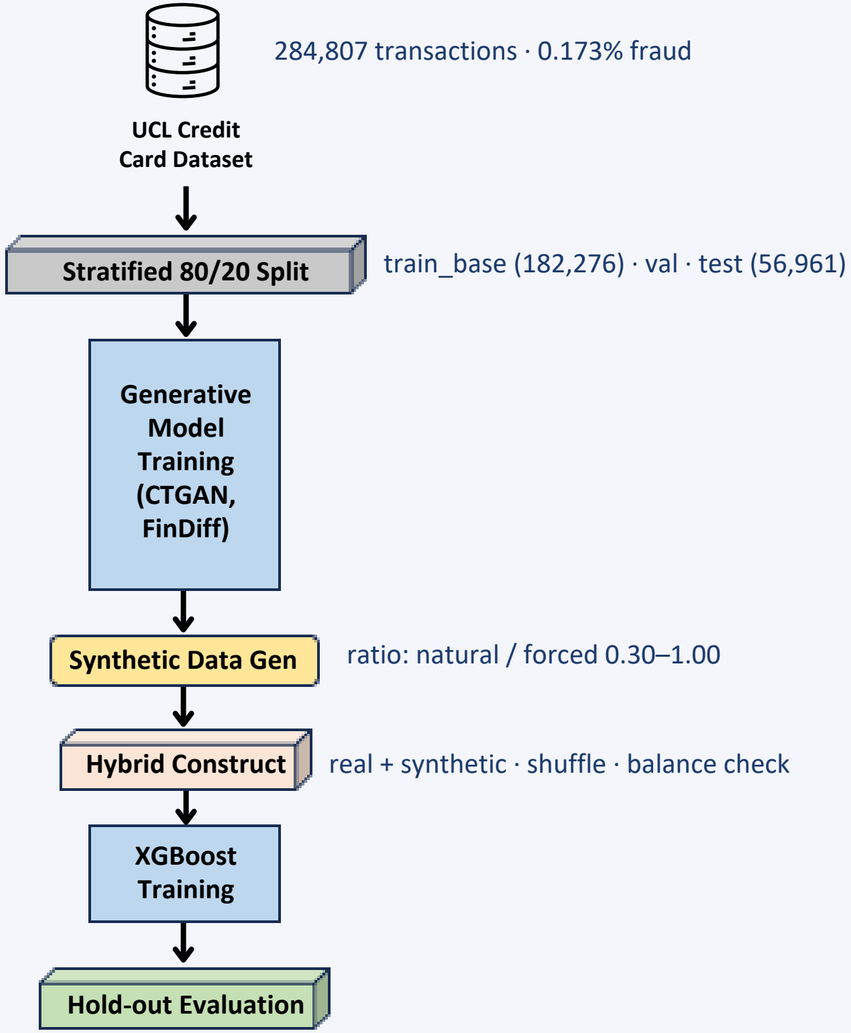
**0.173%**
Fraud rate

**577.9**
scale_pos_weight

**Hold-out split** (80/20 stratified) · **Synthetic size** = len(train_base) = 182,276 · **Scaler:** StandardScaler (QuantileTransformer discarded empirically)

**Data Limitation:** UCI dataset contains only 492 fraud samples, generative models learn from an extremely small minority class, limiting synthetic diversity. PCA-anonymised features (V1–V28) prevent domain-level validation of generated fraud patterns.

## Pipeline Flowchart



METHODOLOGY

# Models, Ratio Control & Evaluation

## CTGAN (GAN-based)

epochs=300, batch=500, embed_dim=128
gen/disc_dim=(256,256), dsteps=1, log_freq=True
Ratio control: condition_column rejection sampling
→ Class sampled independently, then concatenated

## FinDiff (Diffusion-based)

MLP [1024×4], epochs=500, batch=512, lr=1e-4
T=500 steps, linear β-schedule, StandardScaler
Ratio control: label array pre-assigned before sampling
→ Labels injected at every denoising step

## Evaluation Strategy

| | |
|---|---|
| **Training Modes:** | Hybrid (real + syn) and Syn-only |
| **spw Conditions:** | spw ON (577.9) and spw OFF |
| **Ratio Settings:** | natural, 0.30, 0.50, 0.70, 1.00 |
| **Primary Metrics:** | Fraud F1 · AUPRC |
| **Secondary:** | ROC-AUC · KS · Wasserstein |
| **Privacy:** | NN distance · NearDup rate |

Downstream classifier: XGBoost (fixed across ALL experiments)

# Model Selection: Why CTGAN & FinDiff?

## Selection Criteria

### ① Survey coverage

Identified via Shi et al. (2025) — a systematic survey of synthetic tabular data generation covering GAN, Diffusion and LLM-based methods (arXiv:2504.16506). CTGAN listed under Conditional GAN; FinDiff listed under Financial Diffusion.

### ② Reproducible code

Most prior work evaluates fixed pre-generated samples — reproducible generation code is rarely released. CTGAN and FinDiff are among the few models with fully open pipelines, enabling the ratio-forcing experiments that are central to this study.

### ③ Architectural contrast

CTGAN (GAN-based, adversarial training) vs FinDiff (Diffusion-based, iterative denoising) — selected deliberately to compare two fundamentally different generative paradigms under identical conditions.

### References

[1] Xu et al. (2019). Modeling Tabular Data using Conditional GAN. NeurIPS 2019. https://proceedings.neurips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html

[2] Nüßgen et al. (2023). FinDiff: Diffusion Models for Financial Tabular Data Generation. ICAIF 2023. https://dl.acm.org/doi/pdf/10.1145/3604237.3626876

[3] Shi et al. (2025). A Comprehensive Survey of Synthetic Tabular Data Generation. arXiv:2504.16506.

# Data Reproducibility & Hybrid Dataset Construction

## How We Reproduce from Prior Data

**1 Public dataset**

UCI Credit Card Fraud (Kaggle: mlg-ulb/creditcardfraud). Identical to datasets used in Xu et al. (2019) and Nüßgen et al. (2023), enabling direct comparison.

**2 Fixed random seed**

All splits use stratified 80/20 with random_state=42. train_base / val / test are fixed across all 50 experiments.

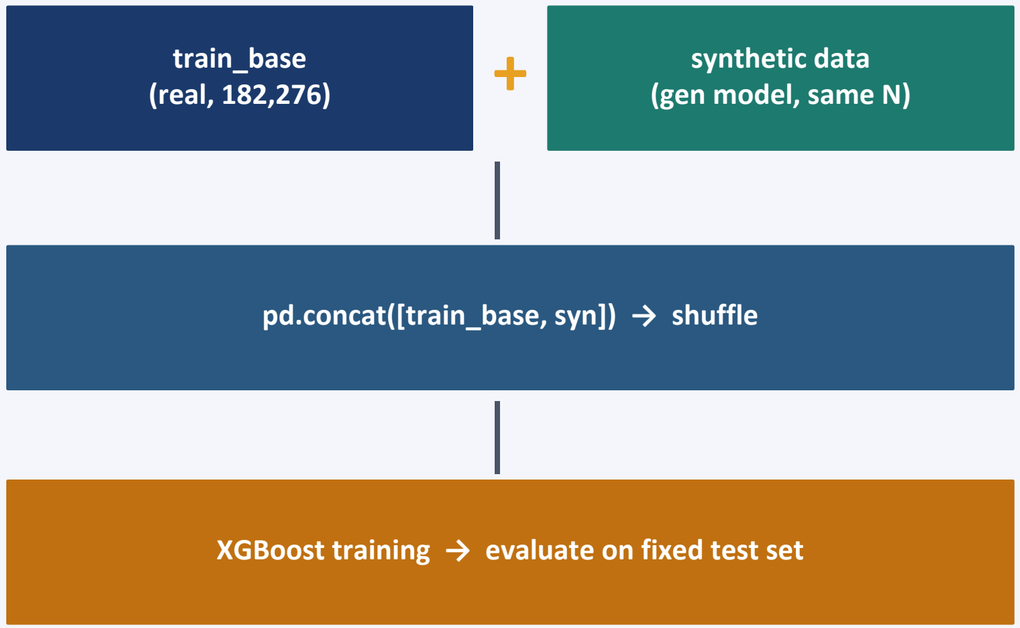
**3 Official model code**

CTGAN via SDV library (sdv-dev/CTGAN). FinDiff from nüßgen/FinDiff GitHub. No modifications to core architecture.

**4 Identical evaluation**

Single fixed hold-out test set (56,961 rows) used for all experiments. XGBoost hyperparameters held constant across all conditions.

## Hybrid Dataset Construction




- Syn size = len(train_base) = 182,276 in all experiments
- For syn-only: train_base replaced entirely by synthetic data
- Fraud ratio in synthetic controlled via ratio parameter (natural or forced)
- StandardScaler fitted on train_base only — no leakage into val/test

*Code: make_hybrid() in ConfusionMat_GAN_.ipynb Cell 17 · find_best_threshold() in param_findiff_.ipynb Cell 10*

RQ1 RESULTS

# Can Synthetic Data Improve or Replace Real Data?

| Configuration | ROC-AUC | AUPRC | Fraud F1 (F1) | Train Data |
|---|---|---|---|---|
| **Real-only – Hold-out (spw ON)** | 0.9684 | 0.8787 | — | Real (227,846) |
| **Real-only – 5-Fold CV avg (spw ON)** | 0.9777 | 0.8608 | **0.866** | Real (80% folds) |
| **FinDiff natural – hybrid (spw OFF) ★** | 0.9780 | 0.8730 | **0.860(0.873)** | Real+Syn (364k) |
| **CTGAN forced_0.30 – hybrid (spw OFF) ★** | 0.9828 | 0.8412 | **0.832(0.829)** | Real+Syn (364k) |
| FinDiff natural – syn-only (spw OFF) | 0.9633 | 0.7247 | 0.774(0.790) | Syn only (182k) |
| CTGAN forced_0.30 – syn-only (spw OFF) | 0.9780 | 0.7034 | 0.745(0.659) | Syn only (182k) |

**Augmentation:** *FinDiff* natural hybrid **slightly outperforms the real-only baseline while using fewer real samples**, showing that **synthetic augmentation can improve fraud detection**. *CTGAN* forced hybrid approaches but **does not exceed the baseline**.

**Replacement:** Syn-only viable only under natural ratio — fragile

**Limitation:** real-only baseline used a larger training set (227,846 vs 182,276); a size-controlled comparison was not conducted.

| Configuration | NN_Mean/Min | NearDup | Fraud_NN_ Mean/Min | KS/Wass |
|---|---|---|---|---|
| **CTGAN syn_only_Natural (spw OFF)** | 89.44/3.67 | 0 | 605.5/10.1 | 0.13/177.8 |
| **CTGAN syn_only_0.5 (spw OFF)** | 82.98/3.70 | 0 | 554.5/7.6 | 0.16/116.1 |
| **FINDIFF syn_only_Natural (spw OFF)** | 222.09/2.33 | 0 | 55659.5/14 | 0.01/68.0 |
| **FINDIFF syn_only_0.5 (spw OFF)** | 24911/3.29 | 0 | 52862/14 | 0.2/939.3 |

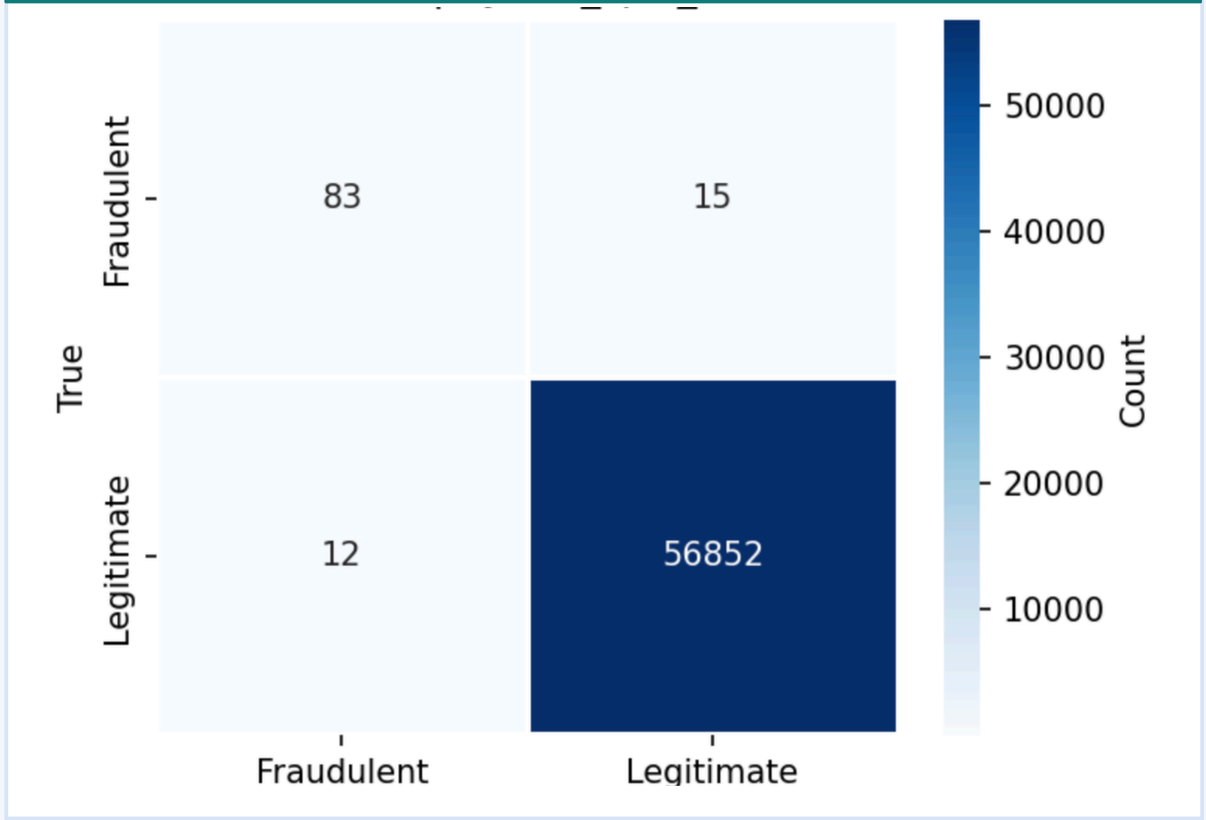
**Privacy/safety Perspective:**
No near-duplicates → no memorisation.
FinDiff natural provides the best balance of realism and privacy.

RQ1 EVIDENCE

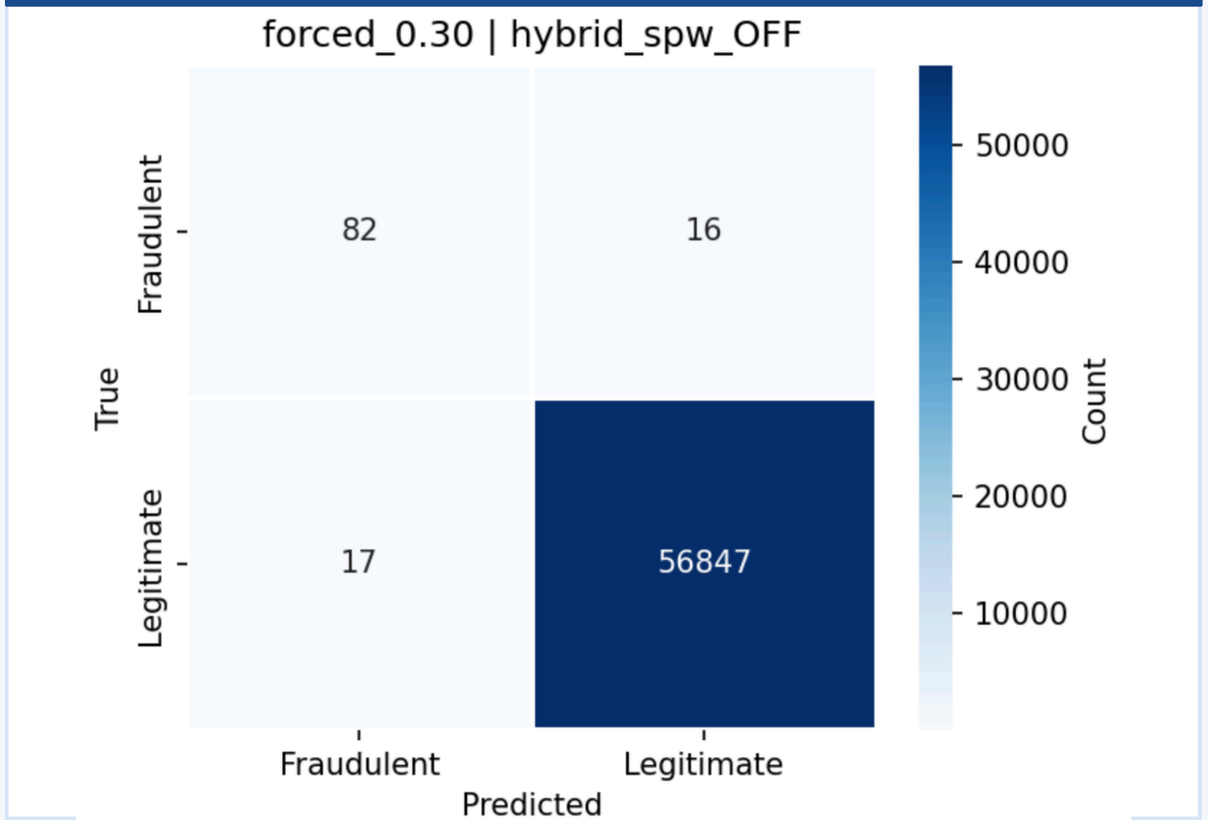
# Confusion Matrices: Best Configurations

## Confusion matrix for best configurations

**FinDiff natural – hybrid** **Fraud F1=0.860 Legit F1=0.999**




**CTGAN 0.3 – hybrid** **Fraud F1=0.8325 Legit F1=0.9997**

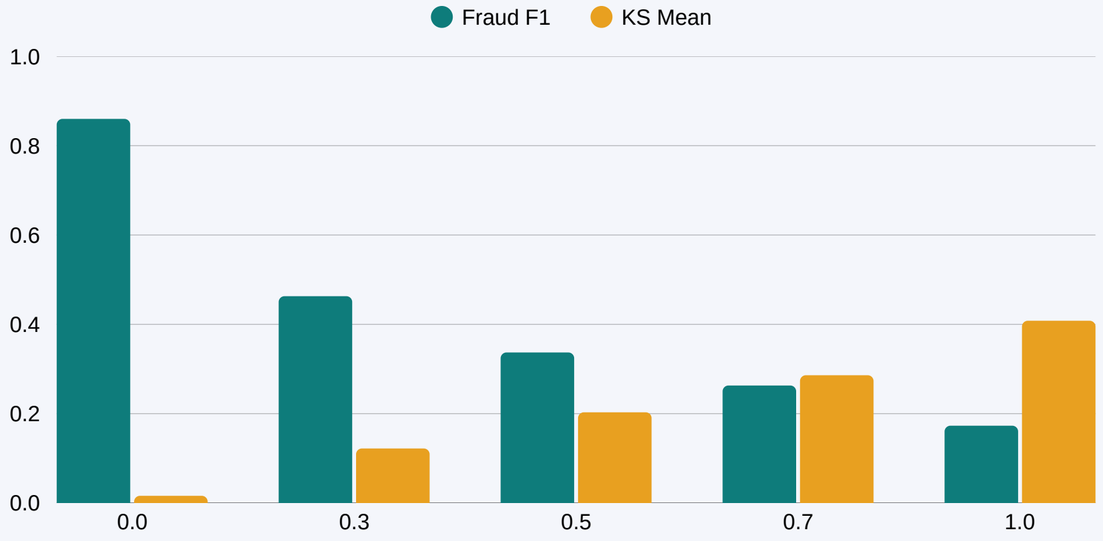



Syn-only best: FinDiff F1=0.774 | CTGAN F1=0.745

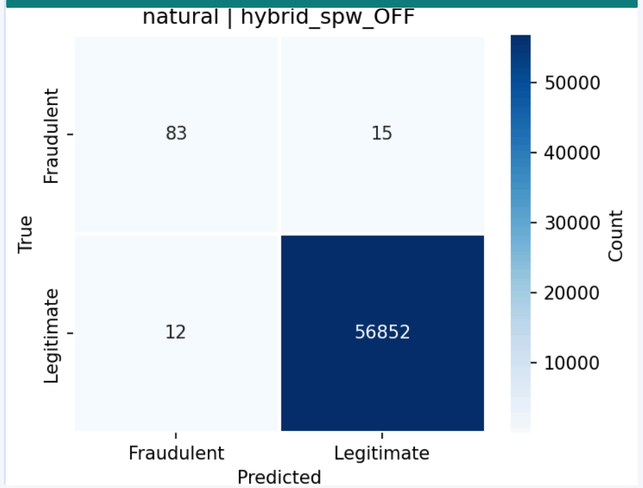
RQ2 RESULTS

What happens when we force different fraud ratios during synthetic data generation?

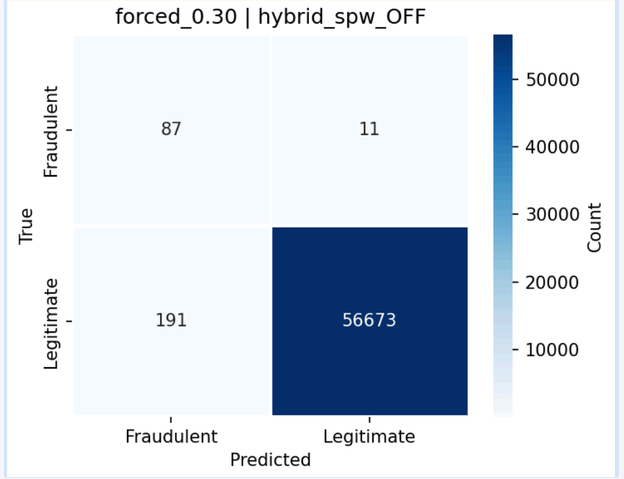
# Ratio Forcing: FinDiff — Gradually decreasing




**natural – F1=0.860 ★** **F1=0.860**




**forced_0.30 – degraded** **F1=0.463**




### Strict Monotonic Degradation

Every ratio increase degrades BOTH KS fidelity and Fraud F1 — no exceptions. KS: 0.016 → 0.408; F1: 0.860 → 0.173

### Wasserstein Amplifies Signal

Distance: 68 (natural) → 1,888 (forced_1.00) — large feature-space magnitude shifts beyond proportion change alone

### Why? Label Conditioning Failure

With label information added at every denoising step, forcing 30% fraud generation from only 315 real fraud samples (0.17%) pushes the model beyond learned fraud patterns.

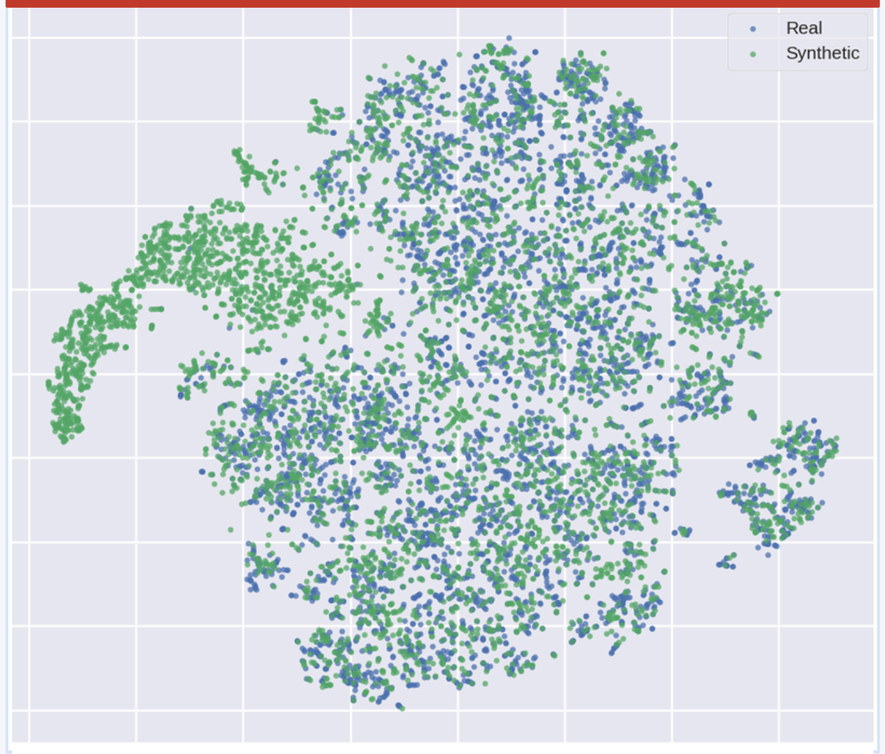
DISTRIBUTION SHIFT

# t-SNE: Fraud Class Distribution — Natural vs Forced

**FINDIFF t-SNE (Real fraud=blue, Synthetic fraud=green)**

**Natural ratio — fraud clusters overlap ✓**

**Forced 0.50 — mode concentration, syn spread out ✗**




**Natural ratio:**
Synthetic fraud overlaps with the real fraud distribution. FinDiff reproduces realistic fraud patterns, allowing the classifier to learn a correct decision boundary.

**Forced ratio:**
Synthetic samples cluster in a narrow region, real fraud becomes underrepresented, and the classifier learns a biased boundary — leading to the F1 drop.

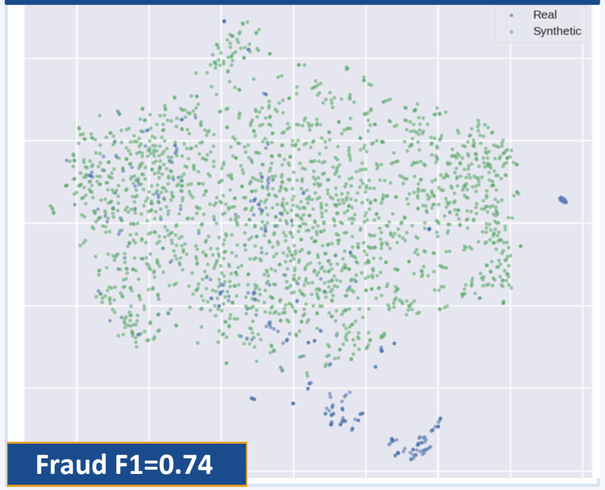
**Limitation:** ratio forcing tested on one dataset only; generalisability to other fraud distributions not confirmed.

RQ2 RESULTS

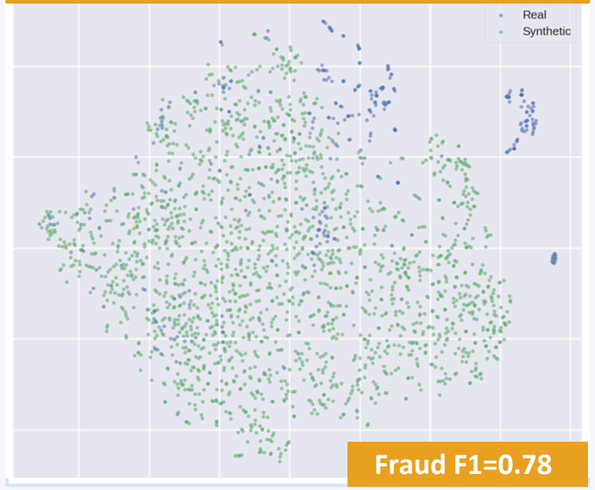
# Ratio Forcing: CTGAN — Irregular & Unexpected Results

**t-SNE: Real vs Synthetic (Real=blue, Synthetic=green)**

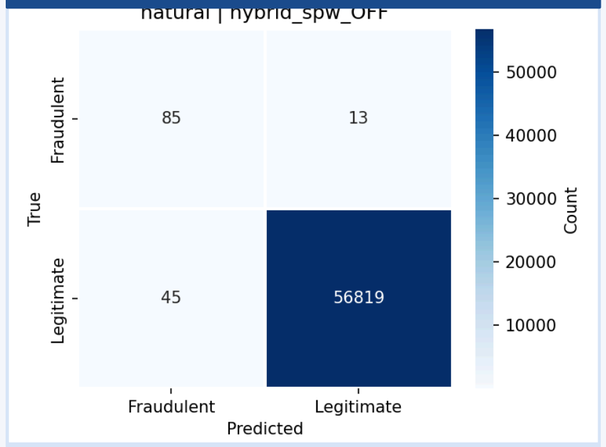
**CTGAN natural (32% fraud)**



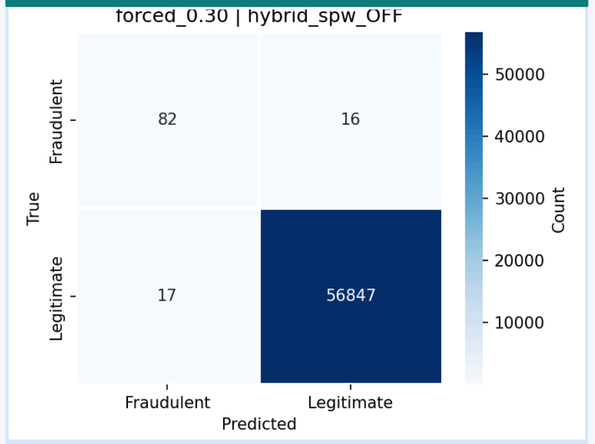

**CTGAN forced_0.50 (54% fraud)**




**natural hybrid (32%) Fraud F1=0.746**




**forced_0.30 hybrid (11%) Fraud F1=0.832**




## Rejection Sampling Failure

condition_column filters generated batches. GAN bias (≈32% fraud):

- forced_0.30 delivers only 11% fraud
- forced_0.70 delivers 96.5% fraud

## Paradoxical Effect

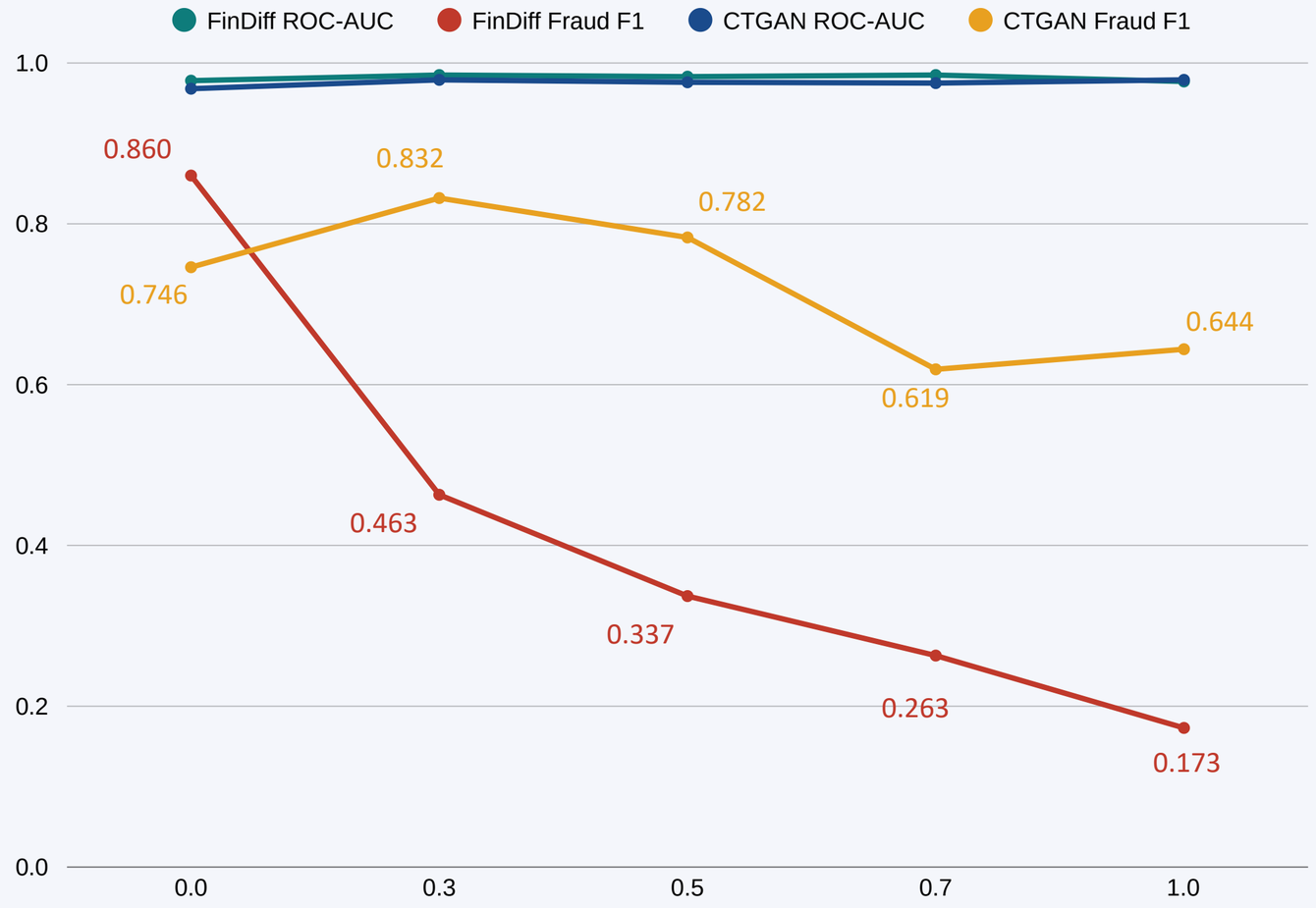
CTGAN's FAILURE to achieve target ratios protects distribution fidelity, actual output stays near natural distribution, limiting distortion

## Mode Collapse Signal

Gen_Loss_Var: 0.765 → 0.8 (forced_0.3) → 0.21 (forced_0.5) → 0.049 (forced_0.70). GAN instability onset. F1 still rises till forced_0.5 - decoupled from generation quality

KEY FINDING

# ROC-AUC Stable, Fraud F1 Collapses — Why?




## Mechanism

### ROC-AUC measures

Ranking ability helps preserved even with skewed training. Feature separation survives moderate distortion.

### Fraud F1 measures

Decision boundary precision is corrupted when synthetic fraud occupies wrong feature regions.

### Implication

ROC-AUC alone is misleading. AUPRC and Fraud F1 are the correct primary metrics for augmentation studies under imbalance.

RQ3 RESULTS

Do GAN and diffusion respond differently to ratio control?

# GAN vs. Diffusion: Opposite Strengths

| Ratio | CTGAN KS | Diff KS | CTGAN F1 | Diff F1 | KS Winner | F1 Winner |
|---|---|---|---|---|---|---|
| **natural** | 0.134 | 0.016 | 0.746 | **0.860** | Diffusion ✓ | Diffusion ✓ |
| forced 0.30 | 0.132 | 0.122 | **0.832** | 0.463 | ≈ Tie | CTGAN ✓ |
| forced 0.50 | 0.165 | 0.203 | **0.783** | 0.337 | CTGAN ✓ | CTGAN ✓ |
| forced 0.70 | 0.227 | 0.286 | **0.619** | 0.263 | CTGAN ✓ | CTGAN ✓ |
| forced 1.00 | 0.378 | 0.408 | **0.644** | 0.173 | CTGAN ✓ | CTGAN ✓ |

### Natural Ratio: Diffusion wins

KS 8× better (0.016 vs 0.134). Deep trajectory conditioning → faithful generation under training distribution

### Forced Ratio: CTGAN more robust

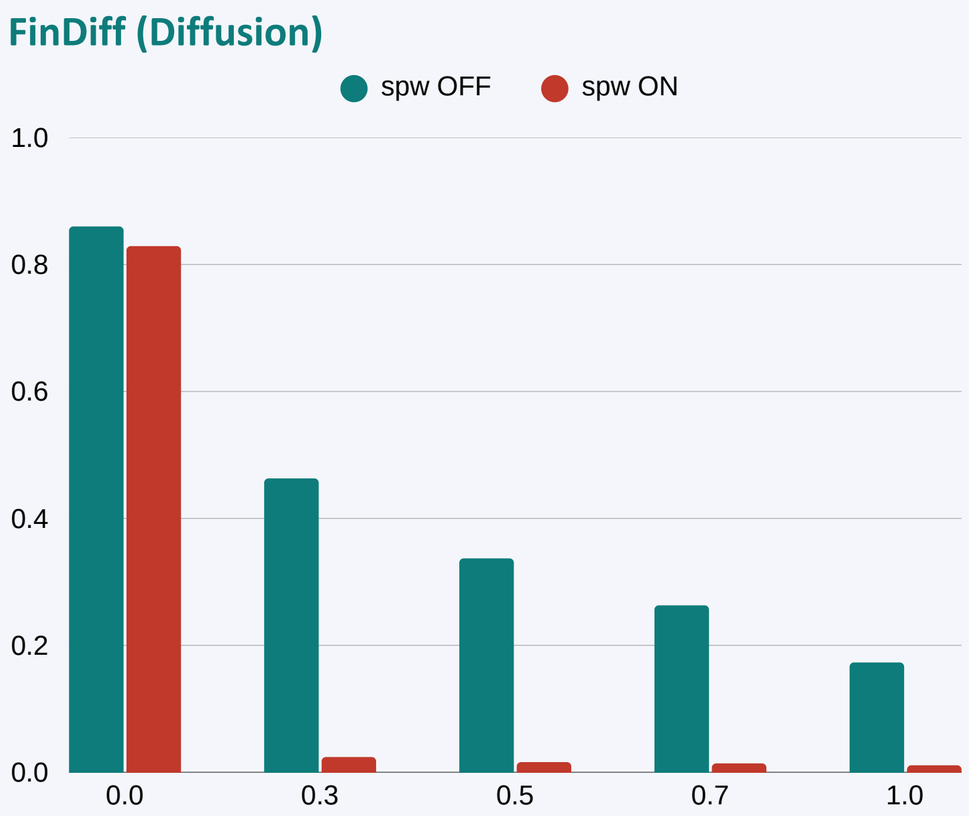
Not because CTGAN generates better — but because it FAILS to achieve target ratios, accidentally preserving fidelity

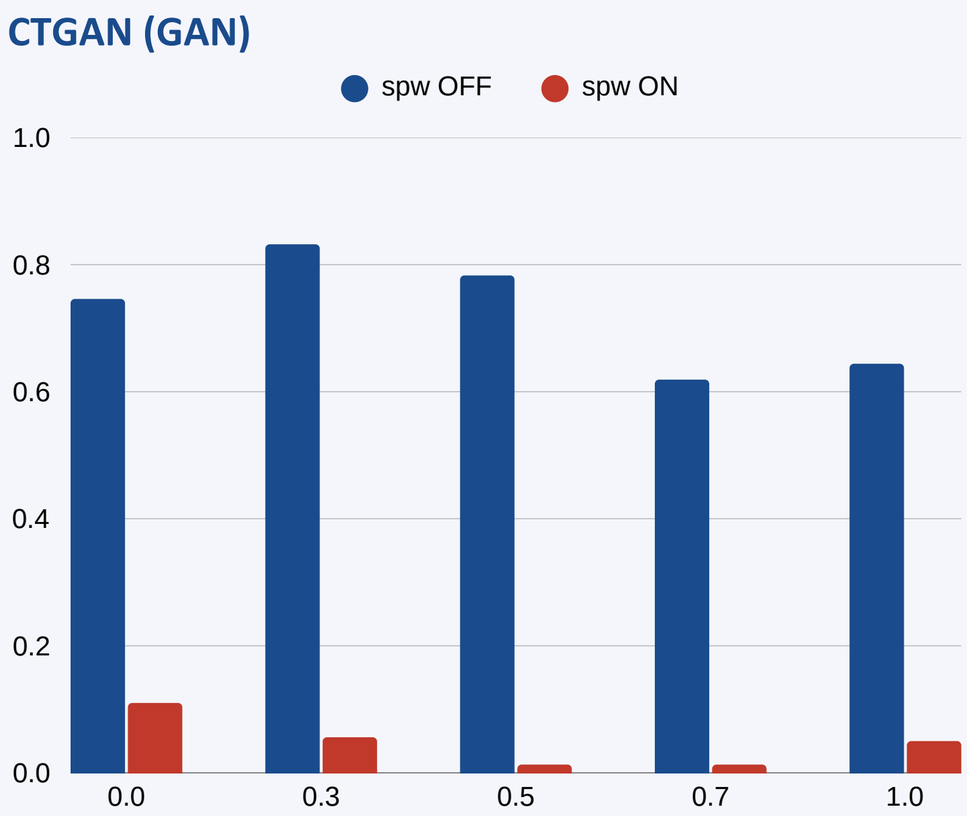
### Converge at matched KS

Similar KS ≠ Similar F1.
CTGAN preserves the boundary by under-generating fraud, while strict ratio control in FinDiff distorts it.

ABLATION STUDY

# Double Penalisation: Forced Ratio + scale_pos_weight




**Key pattern:** natural ratio → spw ON causes moderate damage **any forced ratio → spw ON causes catastrophic collapse (−85% to −98%)**

ABLATION STUDY

# Ablation: Preprocessing & Hyperparameter Sensitivity

### StandardScaler vs QuantileTransformer (FinDiff)

| Metric | StandardScaler | QuantileTransformer |
|---|---|---|
| Train Loss | **0.119** | 0.141 |
| KS Mean | 0.017 | 0.017 |
| Fraud F1 | **0.874** | 0.851 |
| AUPRC | **0.873** | 0.855 |

**QuantileTransformer** turns values into ranks and loses how extreme they are. This makes fraud harder to detect.
**StandardScaler** only rescales the data. Extreme fraud values stay extreme, so the fraud signal remains.

### FinDiff Hyperparameter Sweep (F1 range)

| Parameter | Range Tested | F1 Range |
|---|---|---|
| Epochs | 100 / 300 / 500 | 0.843–0.878 |
| MLP layers | $[256^2]$ / $[512^3]$ / $[1024^4]$ | 0.865–0.877 |
| Diffusion steps | 200 / 500 / 800 | 0.868–0.878 |
| Batch size | 256 / 512 / 768 | 0.860–0.879 |

FinDiff F1 is highly stable (0.843–0.879) across all hyperparameter combinations. Architecture is not the sensitivity but ratio is.

### CTGAN log_frequency Anomaly

log_frequency=False collapses fraud generation to 0.17% (natural) → F1=0.874 (best CTGAN result). CTGAN best performance mimics FinDiff natural behaviour, confirming natural ratio superiority.

### CTGAN log_frequency Anomaly

log_frequency=False collapses fraud generation to 0.17% (natural) → F1=0.874 (best CTGAN result). CTGAN best performance mimics FinDiff natural behaviour, confirming natural ratio superiority.

ABLATION STUDY

# Generalization Study: Does FinDiff Hold on a Different Dataset?

**Why this experiment?**
FinDiff for tabular data remains understudied, prior work has not established whether diffusion-based tabular generators generalise across domain shifts. **Tested on a second dataset (Taiwan Default Payment, 30k samples, 22% default, raw features)** to check whether the findings from the highly imbalanced fraud setting hold more broadly, and how FinDiff compares against SMOTE and TVAE as additional baselines.

**Dataset 1 — UCI Credit Card Fraud (0.17% fraud, 285k samples)**

| Model | Fraud F1 | vs Real | ROC-AUC | AUPRC |
|---|---|---|---|---|
| **Real-only** | 0.860 | *baseline* | 0.993 | 0.869 |
| **SMOTE hybrid** | **0.905** | **+4.5%** | 0.995 | 0.868 |
| FinDiff hybrid | 0.863 | +0.3% | 0.992 | 0.872 |
| TVAE hybrid | 0.813 | −5.4% | 0.981 | 0.822 |
| CTGAN hybrid | 0.750 | −12.8% | 0.980 | 0.854 |

**Dataset 2 — Taiwan Default Payment (22% default, 30k samples)**

| Model | Default F1 | vs Real | ROC-AUC | AUPRC |
|---|---|---|---|---|
| **Real-only** | 0.443 | *baseline* | 0.747 | 0.501 |
| **FinDiff hybrid** | **0.467** | **+5.5%** | 0.781 | 0.549 |
| TVAE hybrid | 0.453 | +2.3% | 0.770 | 0.527 |
| CTGAN hybrid | 0.446 | +0.7% | 0.775 | 0.536 |
| SMOTE hybrid | 0.445 | +0.5% | 0.749 | 0.505 |

**What this tells us:** FinDiff is the only model that ranks **in the top 2 on both datasets** confirming generalizability beyond the fraud domain. **DS1 features are PCA-transformed**, meaning they are continuous and well-structured Creating new samples between existing points in this space often produces realistic data, which is why SMOTE performs well. In contrast, DS2 contains raw categorical variables such as gender and marital status, where creating values between samples can lead to unrealistic results. Diffusion models instead learn the full data distribution, making them more reliable across both datasets.

# Conclusion

## 01 Natural ratio is optimal

Diffusion-based synthesis is most effective when preserving natural class distributions. Forced minority-class ratios consistently degrade FinDiff.

MDPI Technologies 2025 confirms that methods distorting the original class distribution (SMOTE, ADASYN) introduce artificial patterns that misrepresent the minority class, preserving natural distribution is identified as "a critical advantage for imbalanced classification."

## 02 Augment, don't replace

Hybrid training (real + synthetic) matches real-only baseline. Synthetic-only is viable only under natural ratio and FinDiff.

## 03 Never double-penalise

Forced ratio + scale_pos_weight = catastrophic collapse. Synthetic oversampling and algorithmic reweighting are substitutes, not complements.

## 04 Paradigm choice matters

GAN and diffusion respond via fundamentally different mechanisms. Select model based on whether ratio control is needed.

Future work: (1) size-controlled real vs. hybrid comparison · (2) stronger privacy evaluation (membership inference attack) · (3) LLM-based generators (GReaT, CLLM) under same ratio-forcing protocol

Reference
"Improvement of Bank Fraud Detection Through Synthetic Data Generation with Gaussian Noise," Technologies, vol. 13, no. 4, p. 141, 2025. doi: 10.3390/technologies13040141

# Use of AI & Author Contributions

## How AI was used

### Literature Understanding

Helped interpret and summarise complex papers. Diffusion models, CTGAN architecture, and evaluation metrics (AUPRC, KS distance).

### Code Comprehension

Helped explain the structure of existing model implementations provided in research papers. AI explained code behaviour.

## Independent work

### Research Design

- Formulating Research Question
- Experimental Design & research framework
- Dataset selection
- Design of class ratio manipulation experiments

### Experimental Implementation

- Implementation of ratio manipulation experiments
- Parameter exploration & experimental setup
- Evaluation pipeline construction (with limited AI assistance for debugging specific coding errors while following the methodology described in the literature)

### Result Analyses

- Interpretation of all experimental results